Setup

Get the non-negotiable decisions right once, before a second contributor joins

Check your engine

Confirm dbt v1 vs v2 before anything else, install story differs

```
dbt --version                          Confirms which engine
pip install dbt-snowflake              v1: adapter is pinned
dbt init my_project                    Scaffolds a new project
```

Secrets, never literal

A connection block should never hold a password as plain text

```
my_project:
  target: dev
  outputs:
    dev:
      type: snowflake
      account: "{{ env_var('SF_ACCOUNT') }}"
      user: "{{ env_var('SF_USER') }}"
      password: "{{ env_var('SF_PASSWORD') }}"
      role: transformer
```

Schema strategy

One macro turns three target names into three real environments

```
{% macro generate_schema_name(
    custom_schema_name, node
) %}
  {%- if target.name == 'prod' -%}
    prod
  {%- elif target.name == 'ci' -%}
    ci_pr_{{ env_var('GITHUB_PR_NUMBER') }}
  {%- else -%}
    dev_{{ env_var('DBT_USER') }}
  {%- endif -%}
{% endmacro %}
```

Three environments

dev, ci, and prod stay isolated so builds can't collide

```
dev_{{ env_var('DBT_USER') }}
One schema per developer
ci_pr_{{ env_var('GITHUB_PR_NUMBER') }}
Torn down after the PR
prod
The actual serving schema
```

Git basics

A branch per change, and a commit message that says why

```
git checkout -b feature/x              New branch
git commit -m "Refactor stg_orders"    Say why, not what
git push -u origin feature/x           Push and track
gh pr create                           Open the PR
```

RBAC from day one

Three roles cover every real access pattern, any platform

```
LOADER -> TRANSFORMER -> REPORTER      Role chain
CI/CD service account                  Scoped to its schemas only
Human accounts via SSO                 Interactive dev work
Read-only analytics role               BI tools, never write
```

Version pinning

Pin every version so a minor release can't change behavior under you

```
require-dbt-version: ">=1.8.0"         Pin the engine
packages.yml: exact versions           No ranges for adapters
```

Onboarding

Replace a wiki page with one script a new hire actually runs

```
./bootstrap.sh                         Installs, runs dbt deps, provisions
```

Project layout

Materialize each layer once in dbt_project.yml, not per model

```
models:
  my_project:
    staging:
      +materialized: view
    intermediate:
      +materialized: ephemeral
    marts:
      +materialized: table
```

```
stg_ / int_ / fct_ / dim_              Prefix matches the folder
```